

- RAG Learning

1.4 Powers, Roots, Surds & Standard Form

Pearson Edexcel GCSE Mathematics 1MA1 — Number — spec refs N6–N9 — Papers 1 (non-calculator), 2 and 3 (calculator)

1.4.1 Powers and Roots (N6)

A **power** (or **index**) shows repeated multiplication: $a^n = a \times a \times \cdots \times a$ with n factors. In 2^5 the **base** is 2 and the **index** is 5, so $2^5 = 32$.

A **root** undoes a power: $\sqrt{121} = 11$ because $11^2 = 121$; $\sqrt[3]{125} = 5$ because $5^3 = 125$; $\sqrt[5]{32} = 2$ because $2^5 = 32$.

A positive number has two square roots, e.g. ± 9 for 81, but the symbol $\sqrt{}$ always means the positive one. A negative number has a negative cube root: $\sqrt[3]{-27} = -3$.

Facts to Learn

N6 requires you to recognise powers of 2, 3, 4 and 5, as well as square and cube numbers.

Type	Values
Squares to 15^2	1, 4, 9, 16, 25, 36, 49, 64, 81, 100, 121, 144, 169, 196, 225
Cubes	1, 8, 27, 64, 125 (and $10^3 = 1000$)
Powers of 2	2, 4, 8, 16, 32, 64, 128, 256, 512, 1024
Powers of 3	3, 9, 27, 81, 243
Powers of 4	4, 16, 64, 256

Powers of 5

5, 25, 125, 625

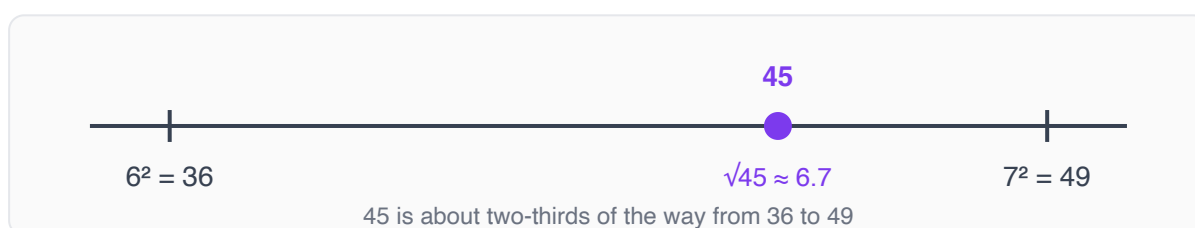
Worked Example: Work out $3^3 \times 2^2 - \sqrt{81}$.

Powers and roots come before multiplication and subtraction: $27 \times 4 - 9 = 108 - 9 = 99$

Estimating Powers and Roots Higher

The 1MA1 Higher version of N6 adds "estimate powers and roots of any given positive number".

Trap the root between two known squares (or cubes), then test decimals.



Worked Example: Estimate $\sqrt{45}$ to 1 decimal place.

$6^2 = 36$ and $7^2 = 49$, so $6 < \sqrt{45} < 7$.

$6.7^2 = 44.89$ and $6.8^2 = 46.24$. 45 is much closer to 44.89, so $\sqrt{45} \approx 6.7$

Worked Example: Estimate 4.1^3 .

$4.1 \approx 4$, so $4.1^3 \approx 4^3 = 64$. This is an underestimate, because $4.1 > 4$ (the true value is 68.921).

1.4.2 Calculating with Indices (N7)

$$a^m \times a^n = a^{m+n} \quad a^m \div a^n = a^{m-n} \quad (a^m)^n = a^{mn}$$

$$a^0 = 1 \quad a^{-n} = \frac{1}{a^n} \quad a^1 = a$$

These laws need the **same base**: $2^3 \times 5^4$ cannot be combined into one power.

Worked Example: Simplify $3^8 \times 3^{-3} \div 3^2$. Give your answer as a power of 3.

$$3^{8+(-3)-2} = \mathbf{3^3} \text{ (which is 27)}$$

Worked Example: Simplify $(5x^2)^3$.

$$\text{Cube the number and the letter part: } 5^3 \times x^{2 \times 3} = \mathbf{125x^6}$$

Zero and Negative Indices

Foundation N7 covers all **integer** indices, so zero and negative powers are on both tiers.

The division law gives $7^2 \div 7^2 = 7^0$, but anything (non-zero) divided by itself is 1, so $7^0 = 1$.

Likewise $7^2 \div 7^5 = 7^{-3}$, while $\frac{49}{16\,807} = \frac{1}{343} = \frac{1}{7^3}$. So a negative index means **one over** (the reciprocal).

Worked Example: Work out (a) 5^{-2} (b) $\left(\frac{2}{5}\right)^{-2}$

$$\text{(a) } \frac{1}{5^2} = \frac{1}{\mathbf{25}} \quad \text{(b) turn the fraction upside down, then square: } \left(\frac{5}{2}\right)^2 = \frac{\mathbf{25}}{4}$$

Worked Example: Find the value of n when $2^n = 32 \times 2^4$.

$$32 = 2^5, \text{ so } 2^n = 2^5 \times 2^4 = 2^9, \text{ giving } \mathbf{n = 9}$$

Exam Tip: A negative power gives a **small positive** answer, not a negative one: $5^{-2} = \frac{1}{25}$, not -25 . Watch brackets too: $(-3)^2 = 9$ but $-3^2 = -9$.

Fractional Indices Higher

Higher N7 extends this to "integer and fractional indices".

$$a^{\frac{1}{n}} = \sqrt[n]{a} \qquad a^{\frac{m}{n}} = \left(\sqrt[n]{a}\right)^m \qquad a^{-\frac{m}{n}} = \frac{1}{\left(\sqrt[n]{a}\right)^m}$$

Bottom of the fraction = **root**; top = **power**; minus sign = **reciprocal**.

Worked Example: Work out $125^{\frac{2}{3}}$.

Root first: $\sqrt[3]{125} = 5$. Then power: $5^2 = \mathbf{25}$

Worked Example: Work out $36^{-\frac{1}{2}}$.

$$\frac{1}{36^{\frac{1}{2}}} = \frac{1}{\sqrt{36}} = \frac{1}{6}$$

Worked Example: Work out $\left(\frac{64}{27}\right)^{-\frac{2}{3}}$.

Reciprocal: $\left(\frac{27}{64}\right)^{\frac{2}{3}}$. Cube root: $\frac{3}{4}$. Square: $\frac{9}{16}$

Worked Example: Solve $4^x = \frac{1}{32}$.

Use base 2 on both sides: $4^x = 2^{2x}$ and $\frac{1}{32} = 2^{-5}$, so $2x = -5$ and $x = -\frac{5}{2}$

Exam Tip: Take the root before the power so the numbers stay manageable without a calculator: $16^{\frac{3}{4}} = 2^3 = 8$, rather than $\sqrt[4]{4096}$.

1.4.3 Calculating Exactly with Fractions and Multiples of π (N8)

"Give your answer in terms of π " means keep π as a symbol rather than using 3.14...; the answer is then **exact**. Exact fractions (such as $\frac{1}{3}$ rather than 0.333) are treated in the same way.

Worked Example: A circle has diameter 18 cm. Work out its exact area and its exact circumference.

$$r = 9: \text{Area} = \pi \times 9^2 = \mathbf{81\pi \text{ cm}^2}; \text{Circumference} = \pi \times 18 = \mathbf{18\pi \text{ cm}}$$

Worked Example: A circle of diameter 8 cm is cut from a square of side 8 cm. Work out the exact area remaining.

Square = 64; circle = $\pi \times 4^2 = 16\pi$. Remaining area = $\mathbf{64 - 16\pi \text{ cm}^2}$ (these are not like terms, so this is the final answer).

Exam Tip: Handle π like a letter: $4\pi + 7\pi = 11\pi$ and $3 \times 5\pi = 15\pi$, but $64 - 16\pi$ cannot be simplified to 48π .

1.4.4 Surds (N8) Higher

Higher N8 adds "calculate exactly with surds; simplify surd expressions involving squares and rationalise denominators".

A **surd** is a root that is irrational — it cannot be written as a fraction — such as $\sqrt{3}$, $\sqrt{7}$ or $\sqrt{20}$. Leaving answers as surds keeps them **exact**. ($\sqrt{16} = 4$ is not a surd.)

$$\sqrt{ab} = \sqrt{a} \times \sqrt{b} \quad \sqrt{\frac{a}{b}} = \frac{\sqrt{a}}{\sqrt{b}} \quad \sqrt{a} \times \sqrt{a} = a$$

Warning: $\sqrt{a+b} \neq \sqrt{a} + \sqrt{b}$ (for example $\sqrt{9+16} = 5$, not 7).

Simplifying Surds

Split the number into the **largest square factor** times what is left.

Worked Example: Simplify $\sqrt{45}$ and $\sqrt{200}$.

$$\sqrt{45} = \sqrt{9 \times 5} = \sqrt{9} \times \sqrt{5} = 3\sqrt{5}$$

$$\sqrt{200} = \sqrt{100 \times 2} = 10\sqrt{2}$$

Worked Example: Simplify $\sqrt{50} + \sqrt{18} - \sqrt{8}$.

$$5\sqrt{2} + 3\sqrt{2} - 2\sqrt{2} = 6\sqrt{2} \quad (\text{like surds are collected like like terms})$$

Expanding Brackets

Worked Example: Expand and simplify $(4 + \sqrt{5})^2$.

$$(4 + \sqrt{5})(4 + \sqrt{5}) = 16 + 4\sqrt{5} + 4\sqrt{5} + 5 = 21 + 8\sqrt{5}$$

Worked Example: Expand and simplify $(6 + \sqrt{7})(6 - \sqrt{7})$.

$$36 - 6\sqrt{7} + 6\sqrt{7} - 7 = 29 \quad (\text{the surd terms cancel: } a^2 - b)$$

Rationalising the Denominator

To **rationalise the denominator** is to rewrite a fraction so that no surd is left on the bottom.

$$\frac{a}{\sqrt{b}} = \frac{a}{\sqrt{b}} \times \frac{\sqrt{b}}{\sqrt{b}} = \frac{a\sqrt{b}}{b}$$

$\frac{a}{c - \sqrt{b}}$: multiply top and bottom by $c + \sqrt{b}$, making the denominator $c^2 - b$

Worked Example: Rationalise the denominator of $\frac{12}{\sqrt{6}}$.

$$\frac{12}{\sqrt{6}} \times \frac{\sqrt{6}}{\sqrt{6}} = \frac{12\sqrt{6}}{6} = 2\sqrt{6}$$

Worked Example: Rationalise the denominator of $\frac{8}{3 - \sqrt{5}}$.

$$\frac{8(3 + \sqrt{5})}{(3 - \sqrt{5})(3 + \sqrt{5})} = \frac{24 + 8\sqrt{5}}{9 - 5} = \frac{24 + 8\sqrt{5}}{4} = 6 + 2\sqrt{5}$$

Exam Tip: "In the form $a\sqrt{b}$ " means fully simplified, with b as small as possible: $\sqrt{48} = 4\sqrt{3}$, not $2\sqrt{12}$.

1.4.5 Standard Form (N9)

A number is in **standard form** when it is written as

$$A \times 10^n \quad \text{where } 1 \leq A < 10 \text{ and } n \text{ is an integer}$$

Numbers of 10 or more have a positive power; numbers between 0 and 1 have a negative power.

Large number

52 000

point moves 4 places

5.2×10^4

Small number

0.00031

point moves 4 places

3.1×10^{-4}

Ordinary number	Standard form
640 000	6.4×10^5
7 300 000	7.3×10^6
0.0009	9×10^{-4}
0.0000481	4.81×10^{-5}

Multiplying and Dividing

Work with the number parts and the powers of 10 separately, then adjust so that $1 \leq A < 10$.

Worked Example: Work out $(8 \times 10^6) \times (7 \times 10^2)$. Give your answer in standard form.

$$8 \times 7 = 56 \text{ and } 10^6 \times 10^2 = 10^8, \text{ giving } 56 \times 10^8.$$

$$56 \text{ is too big for } A: 56 \times 10^8 = 5.6 \times 10^1 \times 10^8 = \mathbf{5.6 \times 10^9}$$

Worked Example: Work out $(6.3 \times 10^4) \div (2.1 \times 10^{-3})$.

$$6.3 \div 2.1 = 3 \text{ and } 10^4 \div 10^{-3} = 10^{4-(-3)} = 10^7, \text{ so the answer is } \mathbf{3 \times 10^7}$$

Adding and Subtracting

You cannot simply add the number parts unless the powers are the same, so write the numbers out in full (or with the same power of 10) first.

Worked Example: Work out $(5.1 \times 10^6) - (2.4 \times 10^5)$.

$$5\,100\,000 - 240\,000 = 4\,860\,000 = \mathbf{4.86 \times 10^6}$$

Calculator Displays

A calculator may show $2.9\text{E-}4$ or 2.9×10^{-4} ; both mean $2.9 \times 10^{-4} = 0.00029$. Always write the answer properly — " $2.9\text{E-}4$ " will not get the mark.

Worked Example: Jupiter is about 7.8×10^8 km from the Sun. Light travels at 3×10^5 km/s. How long does sunlight take to reach Jupiter?

$$\text{time} = \frac{7.8 \times 10^8}{3 \times 10^5} = 2.6 \times 10^3 = 2600 \text{ seconds. } 2600 \div 60 = 43.3 \dots, \text{ so the time is } \mathbf{43 \text{ minutes } 20 \text{ seconds}}$$

Exam Tip: Check your final A: 56×10^8 and 0.56×10^{10} have the right value but are **not** standard form, and Edexcel mark schemes withhold the accuracy mark for them.

Summary

Topic	Key fact
Powers and roots (N6)	Know squares, cubes and powers of 2, 3, 4, 5; estimating roots is Higher
Index laws (N7)	Multiply $\rightarrow$ add indices; divide $\rightarrow$ subtract; power of a power $\rightarrow$ multiply
Zero / negative index (N7)	$a^0 = 1$, $a^{-n} = 1/a^n$
Fractional index (N7, Higher)	$a^{m/n} = (\sqrt[n]{a})^m$

Exact answers (N8)	Leave π (and, at Higher, surds) in the answer
Surds (N8, Higher)	Simplify with square factors; rationalise with $\sqrt{b}$ or the conjugate $c \pm \sqrt{b}$
Standard form (N9)	$A \times 10^n$, $1 \leq A < 10$, n an integer